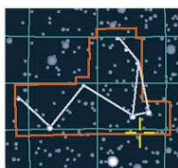


PLANETARY NEBULA

Red Rectangle Nebula



CATALOG NUMBER
HD 44179

DISTANCE FROM SUN
2,300 light-years

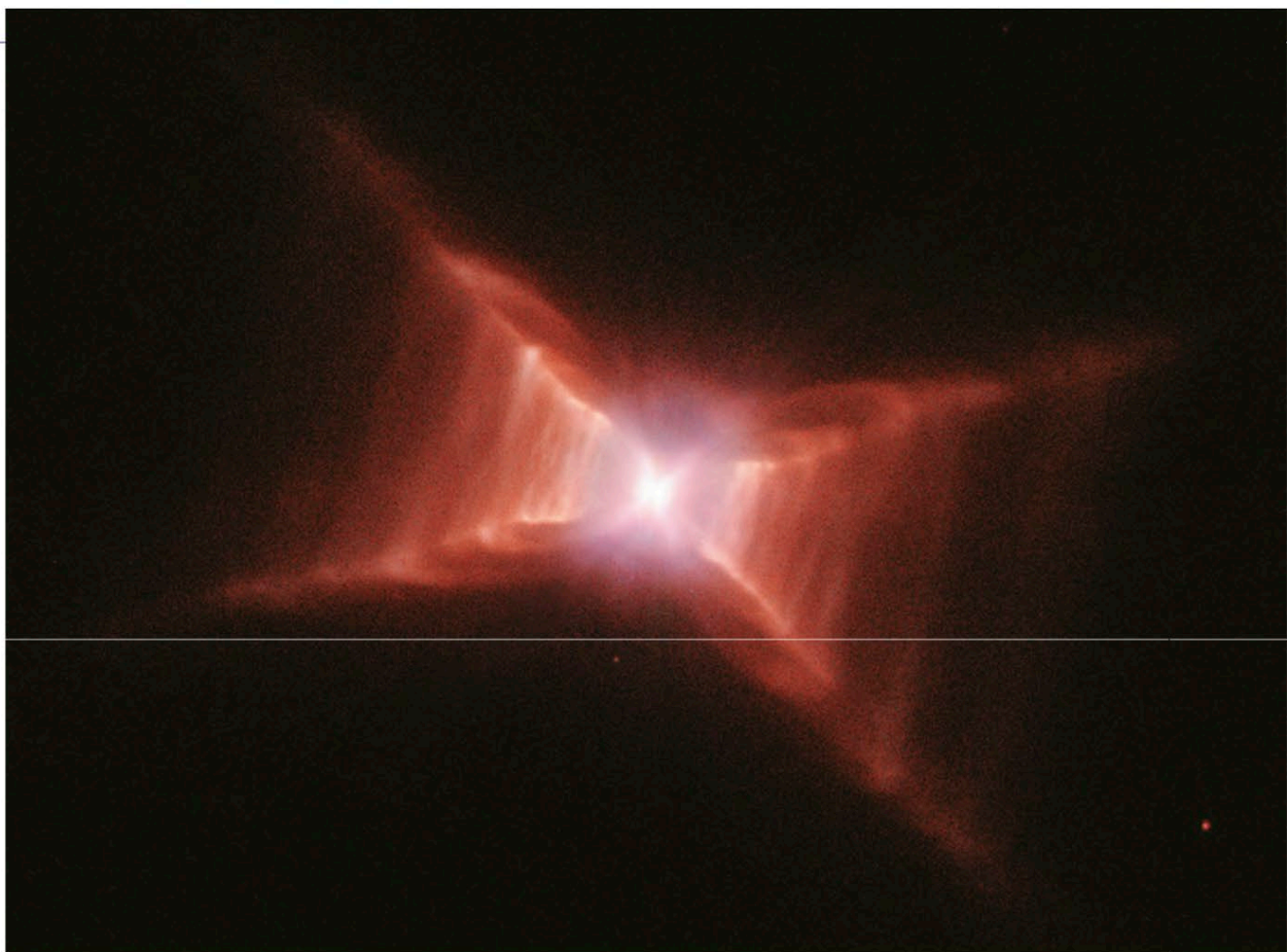
MAGNITUDE 9.02

MONOCEROS

Nature does not often create rectangles, so astronomers were surprised to observe this planetary nebula's unusual shape. The shape of the Red Rectangle Nebula is created by a pair of stars orbiting so close to each other that they experience gravitational interactions. This close binary star has created a dense disk of material around itself, which has restricted the direction of further outflows. This has caused subsequently ejected material to be expelled in expanding cone shapes perpendicular to the disk. Our view of the Red Rectangle is from the side, at right angles to these cones.

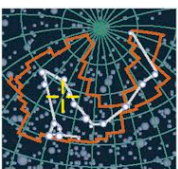
COMPLEX STRUCTURE

One of the most unusual celestial bodies in the Milky Way, the Red Rectangle Nebula has a distinctive shape that reflects an extremely complex inner structure.



PLANETARY NEBULA

Cat's Eye Nebula



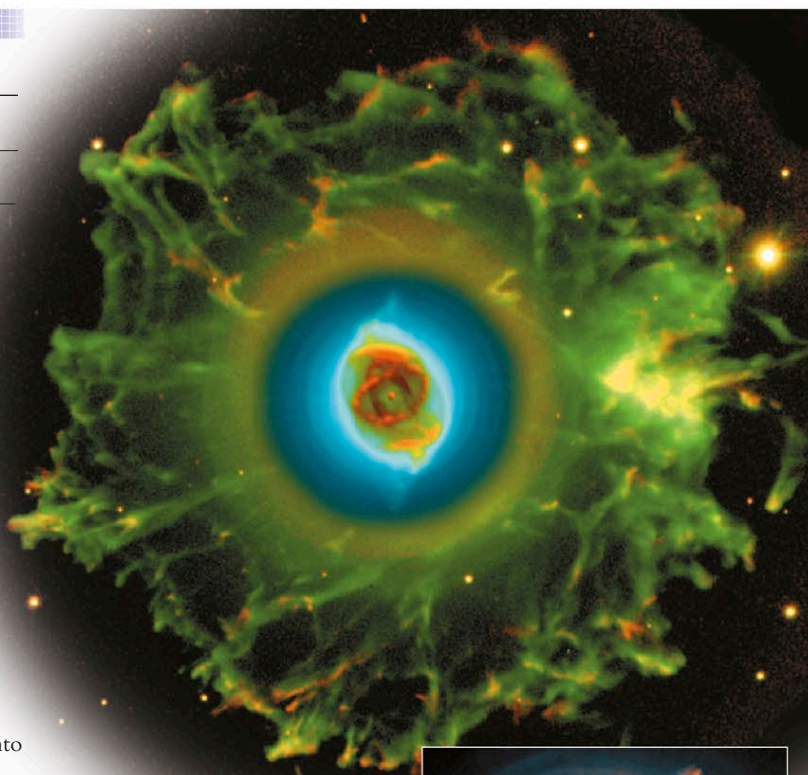
CATALOG NUMBER
NGC 6543

DISTANCE FROM SUN
3,000 light-years

MAGNITUDE 9.8

DRACO

The Cat's Eye Nebula is one of the most complex of all planetary nebulae. It is thought that its intricate structures may be produced either by the interactions of a close binary system or by the recurring magnetic activity of a solitary central star. At 3,000 light-years away, it is too far even for the Hubble Space Telescope to resolve its central star. The "eye" of the nebula is estimated to be more than half a light-year in diameter, with a much larger outer halo stretching into the interstellar medium. Although models of planetary nebulae once assumed a continuous outflow of stellar material, this nebula contains concentric rings that are the edges of bubbles of stellar material ejected at intervals. Eleven of these bubbles have been identified, possibly ejected at intervals of 1,500 years. The Cat's Eye also contains jets of high-speed gas, as well as bow waves created when the gas slammed into slower-moving, previously ejected material.



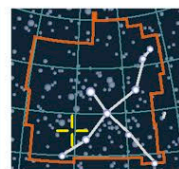
WAVES AND SYMMETRIES

A composite picture (above) shows emission from nitrogen atoms as red and oxygen atoms as green and blue shades, thus revealing successive waves of expelled stellar material. The nebula's symmetrical properties are further revealed by a false-color image processed to highlight its ring structure (right).



PLANETARY NEBULA

Egg Nebula



CATALOG NUMBER
CRL 2688

DISTANCE FROM SUN
3,000 light-years

MAGNITUDE 14

CYGNUS

The Egg Nebula's central star, which was a red giant until a few hundred years ago, is hidden by a dense cocoon of dust (visible in the image below as the dark band of material across the middle of the nebula). The material shed by the dying star is expanding at the rate of 45,000 mph (72,000 kph). Distinct arcs of material suggest a varying density throughout the nebula. The light from the central star shines like searchlights through the thinner parts of its cocoon and reflects off dust particles in the outer layers of the nebula.



BRIGHT SEARCHLIGHTS